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Assessing the impact of early marriage and socioeconomic determinants on under-five morbidity: a cross-country analysis in South Asia

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Abstract

Background Early marriage and socioeconomic factors, which expose young mothers to early pregnancy under situations of adversity, are, as a result, dramatically associated with increased risk of children's morbidity and perpetuate intergenerational cycles of poor health and disparity. Thus, this study aims to assess the association between early marriage and socioeconomic factors on children's morbidity in South Asian countries using national survey data.

Materials and methods This study utilized the most recent nationally representative Demographic and Health Survey (DHS) child datasets from five South Asian countries—Bangladesh, India, Pakistan, Afghanistan, and Nepal—comprising a total sample of 286,131 children. The study's outcome variable was the child morbidity. In addition to descriptive statistics, a two-stage binary logistic regression was used to analyze factors influencing child morbidity.

Results In South Asia, Pakistan had the highest prevalence of child morbidity at 44.38%, followed by Afghanistan at 42.25%, Bangladesh at 34.20%, Nepal at 28.33%, and India with the lowest at 17.72%. Binary logistic regression revealed key factors associated with under-five morbidity in South Asia. Children born to early-married mothers in Pakistan had a significantly higher risk of morbidity (OR = 1.43, 95% CI: 1.20–1.70). Higher morbidity was also associated with maternal secondary education in Pakistan (OR = 1.88, 95% CI: 1.40–2.53), eight or more antenatal care visits in Pakistan (OR = 1.75, 95% CI: 1.17–2.63), Afghanistan (OR = 2.23, 95% CI: 1.83–2.71), and female-headed households in India and Pakistan (OR = 1.14, 95% CI: 1.05–1.24). Breastfeeding was connected to higher child morbidity in Bangladesh, India, Pakistan, and Afghanistan. In contrast, higher maternal education was associated with a significant reduction in child morbidity in both India (OR = 0.81, 95% CI: 0.70–0.93) and Afghanistan (OR = 0.57, 95% CI: 0.45–0.72). Rural residence (OR = 0.80, 95% CI: 0.73–0.87) in Afghanistan, as well as wealth status in India (OR = 0.79, 95% CI: 0.72–0.87), were protective factors.

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Conclusion These findings highlight the urgent need to delay early marriage and address socioeconomic disparities to reduce child morbidity in South Asia. Improving maternal education, enforcing laws to delay the age at marriage, and increasing access to healthcare are crucial for enhancing child health and well-being in the region.

Keywords Early marriage, Under-five morbidity, Child health, Socioeconomic determinants, Binary logistic regression

Introduction

Early marriage, defined as marriage before the age of 18, remains a major social issue globally, particularly in developing countries. In low- and middle-income nations, it disproportionately impacts females and violates their rights. According to UNICEF, around 12 million girls are still married before turning 18 every year. The highest rates are found in South Asia (30%), Latin America (25%), and Sub-Saharan Africa (38%) [1]. According to data from 41 low-income nations, the global rate of early marriage is approximately 30%, with Niger having the highest rate at 75% [2, 3]. In India, 22% of women aged 20 to 24 gave birth before age 18, and 44.5% were married before age 18. Early motherhood is linked to higher risks of stillbirth, neonatal death, low birth weight, and illness [4]. By 2007, 74.9% of Pakistani women aged 20 to 24 had at least one child, with 31.6% giving birth in the first year of their marriage, and 50% being married before turning 18 [5]. In 2018, over one-third of women in Pakistan and Afghanistan were married before turning 18, influenced by conservative Islamic cultural norms [6]. In Nepal, 33% of women aged 20–24 married before age 18, and 8% before 15 years, even though the legal minimum age is 20 years old, or 18 years with parental agreement [7]. Early marriage damages both mothers and their children under the age of five, raising the risk of morbidity. In 2017, over 15,000 children under the age of five died every day worldwide [1]. These risks are associated with adolescent vulnerability, such as maternal sadness, poverty, and hunger. Antenatal care, skillful delivery, and child vaccines are all less common due to limited education and healthcare availability [8]. Young married girls are also at risk because of their shorter birth spacing and higher risk of HIV infection [9]. To determine the influence of early marriage on infant morbidity, it is critical to distinguish between health hazards associated with early childbirth and the socioeconomic determinants and structural vulnerabilities that young mothers experience [5]. Numerous social, demographic, and environmental factors affect children's nutrition and anemia; among these, the mother's age at marriage is a significant predictor of birth outcomes, nutrition, and cognitive development [10]. Respiratory infections and diarrhea are most common in the first two years, affecting growth and increasing child mortality [11]. Approximately 40–60% of children worldwide receive proper care for diarrhea and ARIs, which remain the top killers of children under five [12]. The prevalence of common illnesses, including AIDS, fever,

cough, diarrhea, and malaria, keeps childhood morbidity high [13]. Understanding infection origins and transmission is critical for effective control [14]. Child health is strongly influenced by socioeconomic factors such as parental education, wealth, residence, healthcare access, and maternal nutrition [15]. Effective approaches include teaching and supporting girls, boosting female education quality, providing financial incentives, and engaging parents and community leaders to discourage early marriage [16]. Girls who stay in school longer have a lower chance of getting pregnant [17]. Improving child health and empowering females, while addressing socioeconomic issues, is critical in low-income countries for reducing early marriage and boosting survival and growth [18, 19]. The extant literature highlights the noteworthy influence of early marriage on the rates of morbidity among children in South Asia. Although early marriage and its negative effects on child health have been examined, there remains a significant gap in understanding how early marriage, combined with various socioeconomic factors, specifically affects under-five morbidity in South Asia. This study adds to the existing literature by providing a multi-country comparative analysis of the association between early marriage and under-five morbidity in South Asia using nationally representative DHS data. Unlike many previous studies focusing on single countries, this analysis highlights cross-country heterogeneity and identifies consistent socioeconomic patterns influencing child morbidity, thereby offering broader regional insights for policy and intervention design.

Objectives of the study

The intention of this study is to find out the impact of early maternal marriage and socioeconomic factors on sickness among South Asian children.

The specific objectives are: To assess the prevalence of child illness in several South Asian nations. To mark out the impact of notable socioeconomic factors on morbidity in children. To analyze the effect of early marriage and socioeconomic determinants on child morbidity using statistical models.

Materials and methods

Study design

We used the analysis based on secondary data. These surveys employed a two-stage stratified sampling technique. Data were extracted from the Demographic and Health Surveys (DHS) of Bangladesh, India, Pakistan,

Afghanistan, and Nepal. For each country, the Kids Recode (KR) files were used. Children aged 0–59 months living with their mothers were included. Cases with missing information on morbidity outcomes or key predictors were excluded. Sampling weights were applied to account for the complex survey design. In Bangladesh, data from the 2017–18 BDHS included 8,772 children [20], in India, the 2019–21 NFHS covered 230,870 children [21], in Pakistan, the 2017–18 PDHS included 10,494 children [22], in Afghanistan, the 2015 AfDHS surveyed 31,802 children [23], and in Nepal, the 2022 NDHS involved 5,193 children [24]. The present study only included children under the age of five from these nationally representative household surveys.

Data source

Permission to use the DHS datasets was obtained through a formal request and registration. Data access is granted at: <https://dhsprogram.com/data/available-datasets.cfm>. All data were anonymized before release, and no personal identifiers were available to the researchers.

Target variable

The study's main goal variable is the morbidity indicators of children under five. Child morbidity was defined as the presence of fever, diarrhea, or symptoms of acute respiratory infection (ARI) within two weeks preceding the survey. These indicators were based on maternal reports as recorded in the DHS questionnaires and were coded as binary variables (1 = yes, 0 = no).

Morbidity indicators

Diarrhea in the last two weeks: This was evaluated by asking the kid if they had experienced diarrhea in the previous two weeks, with “1” coding “yes” and “0” coding “no”. Fever within the previous two weeks: This was evaluated by asking if the kid had a fever during that time, with “1” coding “yes” and “0” coding “no”. Two questions measuring whether the kid had experienced a cough and rapid breathing in the previous two weeks were used to determine whether the child had an acute respiratory infection (ARI), with “1” coding “yes” and “0” coding “no”. And then we recoded the morbidity variable with “1” coding “yes” and “0” coding “no”.

Independent variables

The main predictor variable in this study is early marriage, which is defined as an adult marriage when the respondent was 18 years of age or older and an early marriage when the respondent was under 18 [25]. Additionally, several socioeconomic and demographic factors are incorporated into the study to examine their relationship to early marriage, and they serve as controlled variables in the adjusted logistic regression model. There

are several independent variables such as age at first marriage, child sex, multiple births, birth order, maternal age, maternal education, paternal education, mass media exposure, number of ANC visits, delivery by caesarean section, place of delivery, maternal BMI, sex of household head, wealth index, and place of residence.

Statistical analysis

The impact of early marriage and socioeconomic determinants on children's morbidity was analyzed through descriptive statistics and binary logistic regression methods to assess potential risks. The Pearson chi-square test was used to show the association between categorical variables. The binary logistic regression model has also been utilized to assess the association between early marriage and socioeconomic determinants on child morbidity in children under five. Relationships between factors were quantified by computing adjusted odds ratios with 95% confidence intervals. There were controlling variables in adjusted analyses related to child morbidity. Using Microsoft Word, Microsoft Excel, and SPSS Version 25, all data were weighted and examined.

Binary logistic regression model

Binary logistic regression is an appropriate statistical method for modeling the relationship between a dichotomous dependent variable and one or more independent variables, which may be continuous or categorical. In this study, binary logistic regression was employed to estimate the likelihood of the binary outcome based on a set of explanatory variables. The logistic regression model estimates the probability of occurrence of the event of interest by fitting data to a logistic function. The model is defined as:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

where p is the probability of the event, $X_1, X_2, \dots, X_k$ are independent variables, and $\beta_0, \beta_1, \beta_2, \dots, \beta_k$ are the model coefficients estimated through maximum likelihood estimation (MLE). The coefficients are interpreted in terms of odds ratios (OR), where $OR = e^{\beta}$. An odds ratio greater than 1 indicates a positive association with the outcome, whereas an odds ratio less than 1 indicates a negative association.

Results

Prevalence of child morbidity

Showing the child morbidity scenario among children aged under five years across five South Asian countries. Pakistan had the highest percentage, 44.5% of child morbidity in South Asia, while India had the lowest, 17.7%. Of the children from Afghanistan, 42.3% had child

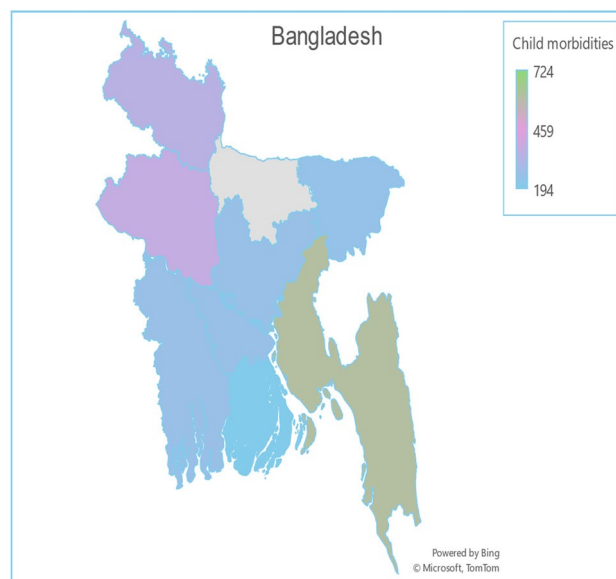
Table 1 Prevalence of child morbidity in South Asian countries

Country	Total individuals	No of illness	Prevalence (%)
Bangladesh	8772	3000	34.2
India	230,870	40,903	17.7
Pakistan	10,494	4665	44.5
Afghanistan	31,802	13,437	42.3
Nepal	5193	1471	28.3

morbidity, followed by Bangladesh at 34.2%, and Nepal at 28.3% as shown in Table 1 and Fig. 1, 2 Other primary information is shown in Table 2 and Figs. 3, 4, 5 and 6.

Association between independent variables and morbidity outcome

The findings of Pearson chi-square tests, which evaluated the bivariate relationships between early maternal marriage, important socioeconomic factors, and the rate of morbidity in children under five, are shown in Table 3. In South Asia, 35.2% of women in Bangladesh, 19.6% in India, and 43.3% in Afghanistan marry before the age of 18, demonstrating that early marriage is common throughout the region. In Bangladesh, 35.3% of male children and 32.9% of female children have morbidity, while in India, the rates are 18.3% for male children and 17.1% for female children. In India, 17.8% of children with morbidity are single births, whereas 13.2% are twins or higher-order multiples. In Pakistan, 44.8% of births are singletons, and 30.7% are multiples. In Afghanistan, 42.4% of births are singletons, and 34.8% are multiples. In Nepal, 28.5% of births are singletons, and 15.2% are multiples. Morbidity affects 17.5% of first-born infants in India, 42.8% of second- or third-born children in Afghanistan, and 29.6% of first-born children in Nepal. The

**Fig. 2** Distribution of child morbidities across eight divisions of Bangladesh

prevalence of morbidity among breastfeeding children is 40.4% in Bangladesh, 21.8% in India, 54.0% in Pakistan, 48.8% in Afghanistan, and 31.4% in Nepal, while among non-breastfeeding children, the rates are 28.7%, 14.3%, 40.5%, 38.9%, and 25.3%, respectively. Morbidity affects 39.3% of children born to women aged 15–18 years in Bangladesh, 25.4% in India, and 55.2% in Pakistan. Child morbidity rates differ by maternal education level: in Bangladesh, 31.6% for uneducated and 30.4% for secondary-educated mothers; in India, 16.7% for uneducated and 15.2% for higher-educated mothers; in Pakistan, 39.9% for uneducated and 51.5% for secondary-educated

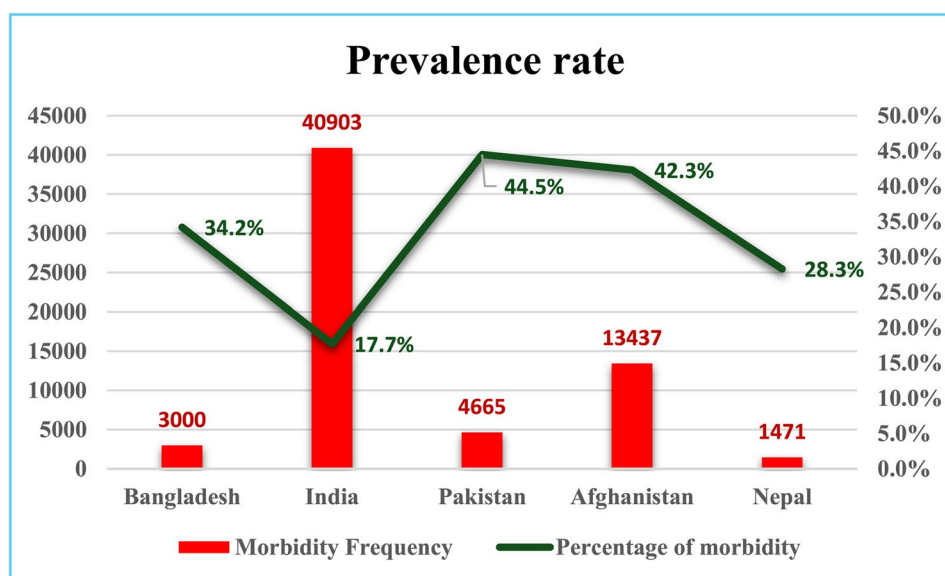
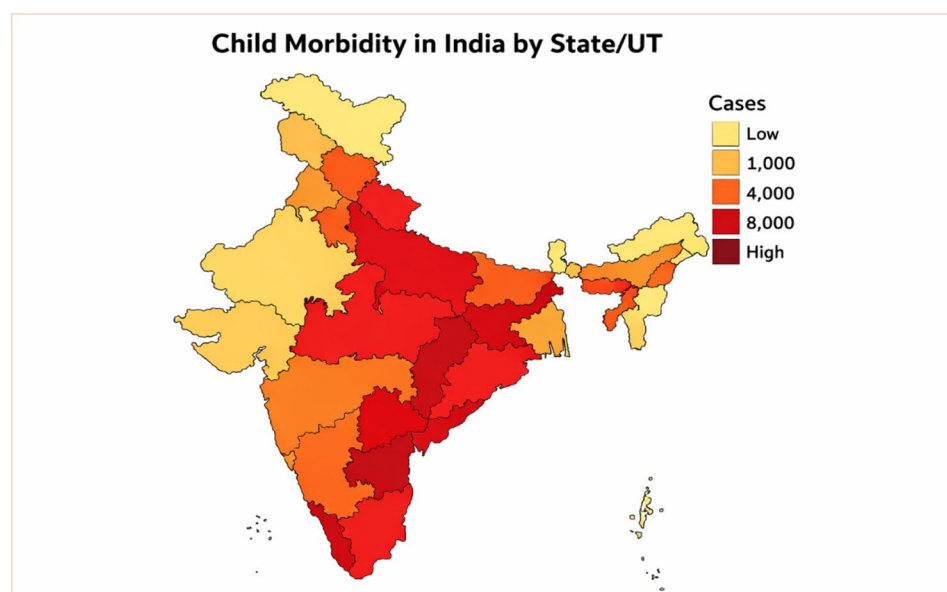
**Fig. 1** Prevalence rate of child morbidity under five in five South Asian countries

Table 2 Frequency distributions and percentage of study participants

Characteristics of Study Participants	Explanatory Variables/ Category	Frequency and Percentage				
		Bangladesh	India	Pakistan	Afghanistan	Nepal
		FrequencyN (%)	FrequencyN (%)	FrequencyN (%)	FrequencyN (%)	FrequencyN (%)
Children characteristics	Age at first marriage					
	Early marriage	6257(71.3)	77275(33.5)	3571(34.0)	16386(51.7)	2494(48.0)
	Adult marriage	2515(28.7)	153195(66.5)	6917(66.0)	15293(48.3)	2697(52.0)
	Child sex					
	Male	4583(52.2)	119960(52.0)	5263(50.2)	16414(51.6)	2729(52.6)
	Female	4189(47.8)	110910(48.0)	5231(49.8)	15387(48.4)	2464(47.4)
	Child twin					
	Single birth	8610(98.1)	226858(98.3)	10240(97.6)	31265(98.3)	5127(98.7)
	Multiple births	163(1.9)	4012(1.7)	254(2.4)	537(1.7)	66(1.3)
	Birth order					
	First	3395(38.7)	90326(39.1)	2548(24.3)	6068(19.2)	2123(40.9)
	Second or third	4274(48.7)	112911(48.9)	4050(38.7)	10364(32.8)	2485(47.9)
	Fourth and up	1104(12.6)	27600(12.0)	3868(37.0)	15190(48.0)	584(11.3)
	Currently breastfeeding					
	No	4634(52.8)	126235(54.7)	7397(70.5)	21097(66.3)	2604(50.1)
	Yes	4138(47.2)	104635(45.3)	3098(29.5)	10704(33.7)	2589(49.9)
Parental characteristics of children	Mother's age					
	15–18	641(7.3)	2404(1.0)	134(1.3)	560(1.8)	148(2.9)
	19–49	8131(92.7)	228466(99.0)	10360(98.7)	31242(98.2)	5045(97.1)
	Maternal education					
	No education	645(7.4)	49306(21.4)	5178(49.3)	26567(52.3)	1153(22.2)
	Primary	2528(28.8)	28434(12.3)	1746(16.6)	2504(7.9)	1830(35.2)
	Secondary	4248(48.4)	117031(50.7)	2251(21.5)	2242(7.0)	1995(38.4)
	Higher	1351(15.4)	36099(15.6)	1319(12.6)	489(1.5)	215(4.1)
	Father's education					
	No education	1301(15.1)	5322(15.1)	3062(29.6)	18300(57.7)	585(11.4)
	Primary	2961(34.3)	4655(13.2)	1793(17.3)	4606(14.5)	2043(39.7)
	Secondary and Higher	4339(50.3)	25033(71.2)	5458(52.8)	8552(27.0)	2412(46.9)
	Don't know	28(0.3)	154(0.4)	28(0.3)	250(0.8)	100(2.0)
	Mass media exposure					
	Not Exposed	3982(45.4)	115139(49.9)	5375(51.2)	15175(47.7)	2855(55.0)
	Exposed	4791(54.6)	115731(50.1)	5119(48.8)	16626(52.3)	2338(45.0)
	Number of ANC					
	No ANC	405(8.0)	12956(7.4)	845(12.6)	7854(40.0)	75(2.6)
	Less than 8	4088(80.9)	128333(73.4)	4929(73.4)	11307(57.6)	2603(91.5)
	8+	559(11.1)	33658(19.2)	938(14.0)	471(2.4)	166(5.9)
	Delivery by caesarean section					
	No	3584(67.2)	181236(78.5)	8143(77.7)	30865(97.3)	2499(81.4)
	Yes	1747(32.8)	49634(21.5)	2339(22.3)	861(2.7)	572(18.6)
	Place of delivery					
	Respondents Home	2670(50.1)	25848(11.2)	3533(33.7)	16188(51.1)	609(19.8)
	With Health Facility	2667(49.9)	205022(88.8)	6958(66.3)	15484(48.9)	24162(80.2)
	Mother's BMI					
	Underweight	1175(14.3)	41522(20.0)	408(12.0)	N/A	346(13.8)
	Normal	4886(59.5)	125295(60.4)	1497(44.2)		1519(60.7)
	Overweight/obese	2154(26.2)	40614(19.6)	1485(43.8)		639(25.5)

Table 2 (continued)

Characteristics of Study Participants	Explanatory Variables/ Category	Frequency and Percentage				
		Bangladesh	India	Pakistan	Afghanistan	Nepal
		FrequencyN (%)	FrequencyN (%)	FrequencyN (%)	FrequencyN (%)	FrequencyN (%)
Household characteristics and Community-level factors	Sex of household head					
	Male	7599(86.6)	195828(84.8)	9334(88.9)	31447(98.9)	3549(68.3)
	Female	1173(13.4)	35040(15.2)	1160(11.1)	354(1.1)	1644(31.7)
	Wealth index					
	Poor	3664(41.8)	106941(46.3)	4486(42.7)	12632(39.7)	2346(45.2)
	Middle	1647(18.8)	45101(19.5)	2178(20.8)	6779(21.3)	1079(20.8)
	Rich	3462(39.4)	78828(34.2)	3831(36.5)	12391(39.0)	1767(34.0)
	Place of residence					
	Urban	2413(27.5)	61528(26.7)	3351(31.9)	7246(22.8)	3367(64.8)
	Rural	6359(72.5)	169342(73.3)	7143(68.1)	24555(77.2)	1826(35.2)

**Fig. 3** Distribution of child morbidities across thirty-seven states of India

mothers; in Afghanistan, 46.1% for primary and 33.5% for higher-educated mothers; and in Nepal, 24.8% for uneducated and 30.4% for secondary-educated mothers. Children with uneducated fathers have a morbidity rate of 16.8% in India, 41.9% in Pakistan, and 43.0% in Afghanistan. 17.4% of mothers in India, 47.0% in Pakistan, and 29.7% in Nepal have access to mass media, which is critical for raising knowledge regarding child health and morbidity. The incidence of child morbidity among mothers with no ANC visits is 23.5% in India, 44.3% in Pakistan, and 40.7% in Afghanistan, but for those with 8 or more ANC visits, the rates are 19.5% in India, 58.1% in Pakistan, and 1.4% in Afghanistan. Morbidity among children born via cesarean section is 18.5% in India, 47.7% in Pakistan, and 49.0% in Afghanistan. In Pakistan, 41.7% of deliveries are normal, and 45.8% take place at health facilities. Child morbidity rates among mothers with an

underweight BMI are 35.9% in Bangladesh and 19.7% in India, compared to 32.4% and 16.7% for mothers with an overweight or obese BMI, respectively. In India, child morbidity rates are 17.4% for male-headed households and 19.4% for female-headed households. Child morbidity rates are 36.0% in Bangladeshi middle-class households, 19.1% in Indian poor households, and 45.8% in Pakistani wealthy households. Child morbidity rates in urban areas of Bangladesh are 31.9%, 16.1% in urban India, 45.1% in urban Afghanistan, and 29.4% in urban Nepal.

Significant risk factors for under-five child morbidity across South Asian countries

To better understand the determinants of under-five morbidity, Table 4 presents the adjusted odds ratios (OR) derived from binary logistic regression models across

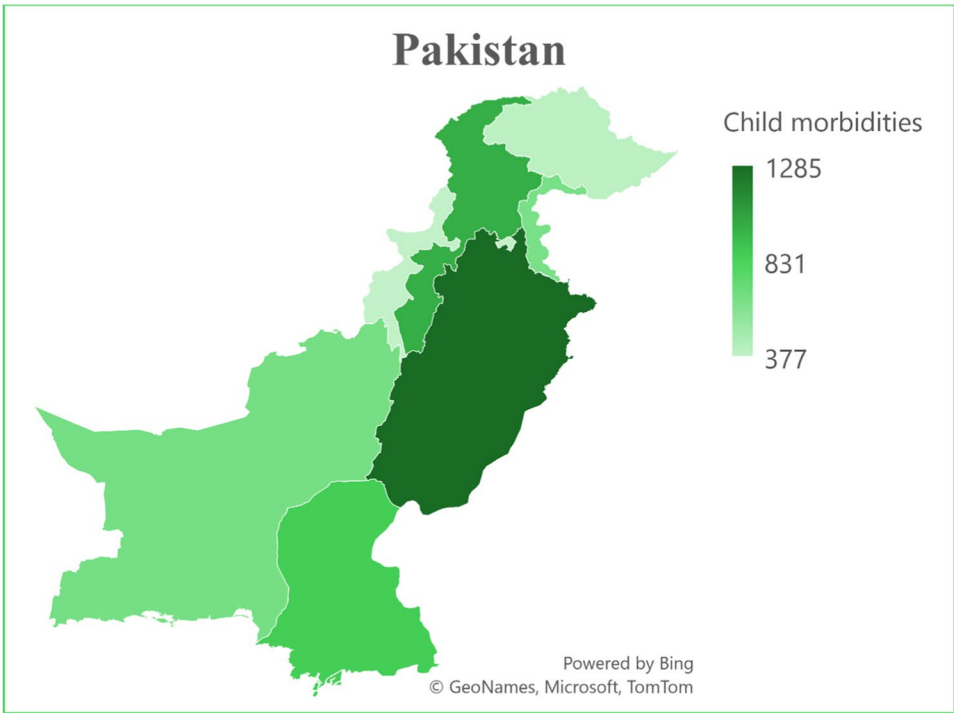


Fig. 4 Distribution of child morbidities across eight states of Pakistan

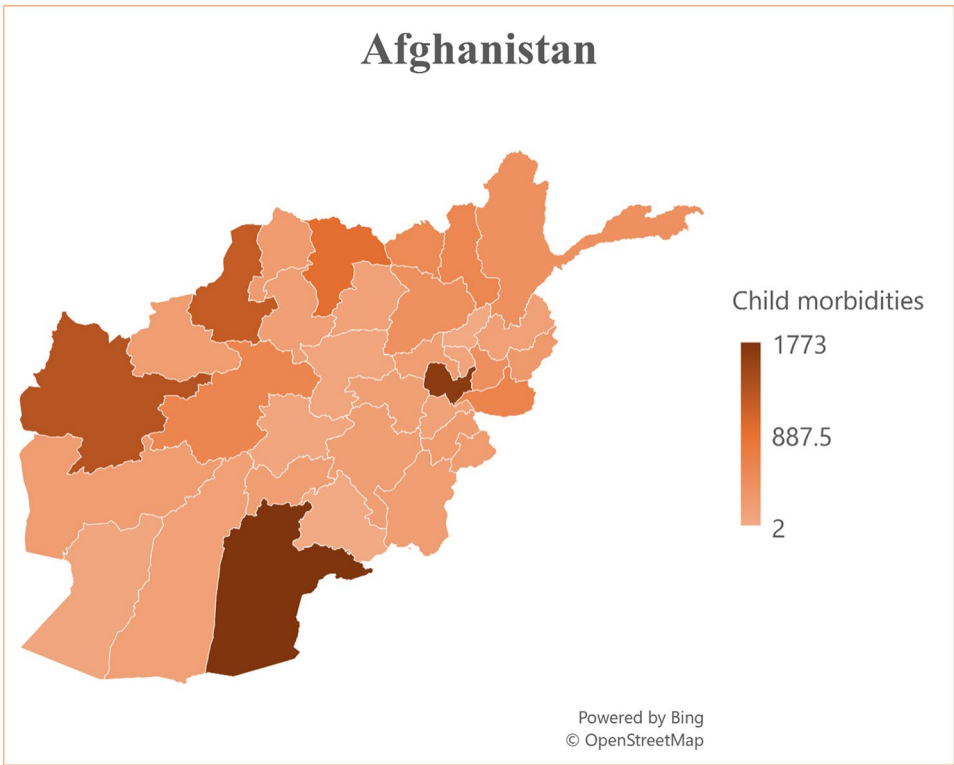


Fig. 5 Distribution of child morbidities across thirty-four states of Afghanistan

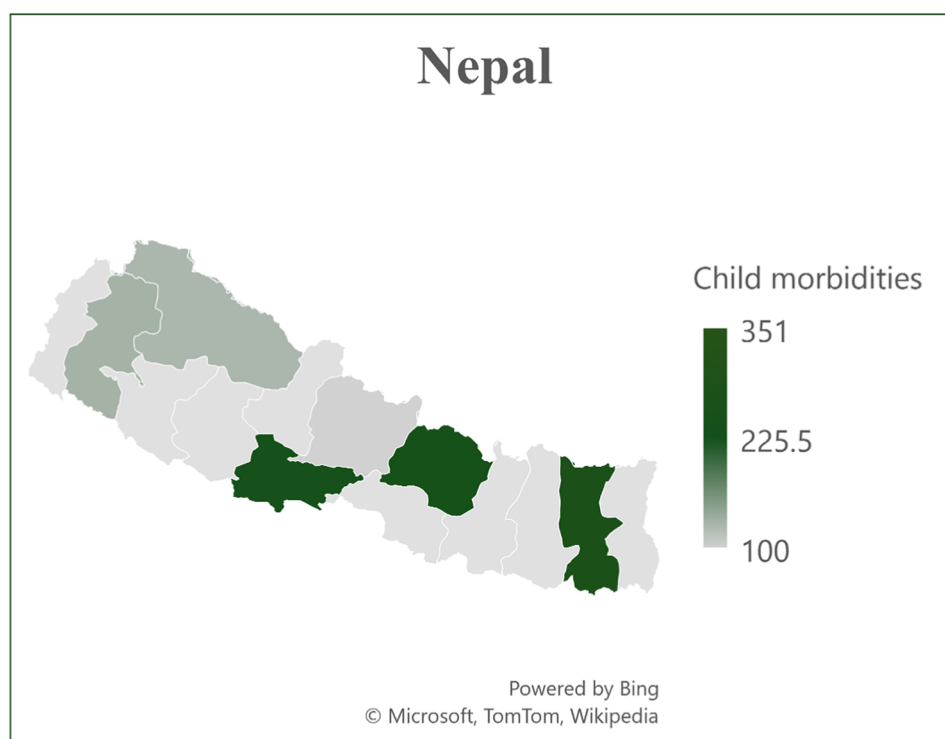


Fig. 6 Distribution of child morbidities across seven provinces of Nepal

five South Asian countries. The study finds that various child, mother, and household-level characteristics are strongly linked with the risk of morbidity. The following is a descriptive overview that focuses on the factors that exhibited a statistically significant positive association with child morbidity, provided by the nation. Breastfeeding was the only significant risk factor in Bangladesh, with children being 29% more likely to suffer morbidity (AOR=1.29, 95% CI: 1.09–1.52), probably due to reverse causation. In India, morbidity was significantly higher among breastfed children (AOR=1.32, 95% CI: 1.24–1.42), children of mothers with secondary education (AOR=1.12, 95% CI: 1.01–1.24), fathers with secondary/higher education (AOR=1.16, 95% CI: 1.04–1.29), those exposed to mass media (AOR=1.08, 95% CI: 1.01–1.16), cesarean births (AOR=1.09, 95% CI: 1.01–1.18), and children from female-headed households (AOR=1.14, 95% CI: 1.05–1.24), and Adult marriage decreased the chance of morbidity (AOR=0.82, 95% CI: 0.77–0.88), as did maternal higher education (AOR=0.81, 95% CI: 0.70–0.93), rural residency (AOR=0.87, 95% CI: 0.80–0.94), and normal maternal BMI. In Pakistan, female children were more likely to have morbidity (AOR=1.43, 95% CI: 1.20–1.70), whereas maternal secondary education (AOR=1.88, 95% CI: 1.40–2.53) and eight or more antenatal care (ANC) visits (AOR=1.75, 95% CI: 1.17–2.63) were also linked with increased risk. In Afghanistan, breastfed children (AOR=1.07, 95% CI: 1.01–1.13),

children of older mothers (19–49 years) (AOR=1.22, 95% CI: 1.01–1.49), and those whose mothers had 8 or more ANC visits (AOR=2.23, 95% CI: 1.83–2.71) had significantly higher morbidity rates, and protective factors against child morbidity included adult marriage (AOR=0.87, 95% CI: 0.82–0.92), multiple births (AOR=0.62, 95% CI: 0.47–0.83), maternal higher education (AOR=0.57, 95% CI: 0.45–0.72), delivery at a health facility (AOR=0.78, 95% CI: 0.73–0.83), and rural residence (AOR=0.80, 95% CI: 0.73–0.87). In Nepal, none of the variables investigated had statistically significant relationships with under-five morbidity.

Discussion

Given the cross-sectional nature of DHS data, this study does not establish causal relationships between early marriage and child morbidity. Therefore, rather than being causative, the observed associations should be understood as correlational. Some of the associations identified in this study may be influenced by unmeasured confounding and reverse causation. Nevertheless, the consistency of associations across multiple countries underscores the importance of early marriage and socioeconomic disadvantage as contextual risk markers for poor child health outcomes in South Asia. The study assessed the association between socio-demographic characteristics and the outcome of child health [26]. Early maternal marriage (< 18 years) shows a significant

Child morbidity

Categories	Bangladesh			India			Pakistan			Afghanistan			Nepal		
	No N (%)	P-value	Yes N (%)	No N (%)	P-value	Yes N (%)	No N (%)	P-value	Yes N (%)	No N (%)	P-value	Yes N (%)	No N (%)	P-value	
Age at first marriage															
Early marriage	2204(35.2)	4053(64.8)	<0.001	15,118(19.6)	62,157(80.4)	<0.001	1567(43.9)	2005(56.1)	0.380	7095(43.3)	9290(56.7)	<0.001	684(27.4)	1810(72.6)	
Adult marriage	796(31.7)	1719(68.3)		25,695(16.8)	127,501(83.2)		3097(44.8)	3821(55.2)		6297(41.2)	8996(58.8)		786(29.1)	1911(70.9)	
Child sex															
Male	1620(35.3)	2963(64.7)	<0.018	21,964(18.3)	97,995(81.7)	<0.001	2355(44.4)	2908(55.3)	0.545	7016(42.7)	9398(57.3)	0.067	800(29.3)	1929(70.7)	
Female	1380(32.9)	2809(67.1)		18,939(17.1)	91,971(82.9)		2310(44.2)	2921(55.8)		6421(41.7)	8966(58.3)		671(27.2)	1793(72.8)	
Child twin															
Single birth	2950(34.3)	5660(65.7)	0.366	40,372(17.8)	186,486(82.2)	<0.001	4587(44.8)	5653(55.2)	<0.001	13,251(42.4)	18,014(57.6)	<0.001	1461(28.5)	3666(71.5)	
Multiple births	50(30.9)	112(69.1)		531(13.2)	3481(86.8)		78(30.7)	176(69.3)		187(34.8)	350(65.2)		10(15.2)	56(84.8)	
Birth order															
First	1112(32.8)	2283(67.2)	0.070	15,803(17.5)	74,524(82.5)	<0.001	1157(45.4)	1391(54.6)	0.439	2449(40.4)	3619(59.6)	<0.005	605(28.5)	1518(71.5)	
Second or third	1495(35.0)	2779(65.0)		19,994(17.7)	92,916(82.3)		1801(44.5)	2250(55.5)		4440(42.8)	5924(57.2)		736(29.6)	1749(70.4)	
Fourth and up	393(35.6)	710(64.4)		5101(18.5)	22,498(81.5)		1693(43.8)	2174(56.2)		6455(42.5)	8736(57.5)		130(22.3)	454(77.7)	
Currently breastfeeding															
No	1328(28.7)	3307(71.3)	<0.001	18,082(14.3)	108,153(85.7)	<0.001	2993(40.5)	4403(59.5)	<0.001	8213(38.9)	12,885(61.1)	<0.001	658(25.3)	1946(74.7)	
Yes	1672(40.4)	2466(59.6)		22,822(21.8)	81,814(78.2)		1672(54.0)	1426(46.0)		5225(48.8)	5479(51.2)		813(31.4)	1776(68.6)	
Mother's age															
15–18	252(39.3)	389(60.7)	<0.005	610(25.4)	1793(74.6)	<0.001	74(55.2)	60(44.8)	<0.012	241(43.1)	318(56.9)	0.678	46(31.1)	102(68.9)	
19–49	2748(33.8)	5383(66.2)		40,293(17.6)	188,173(82.4)		4591(44.3)	5769(55.7)		13,196(42.2)	18,046(57.8)		1425(28.2)	3620(71.8)	
Maternal education															
No education	204(31.6)	441(68.4)	<0.002	8255(16.7)	41,051(83.3)	<0.001	2065(39.9)	3113(60.1)	<0.001	11,168(42.0)	15,399(58.0)	<0.001	286(24.8)	866(75.2)	
Primary	866(34.3)	1661(65.7)		5212(18.3)	32,222(81.7)		832(47.7)	913(52.3)		1154(46.1)	1350(53.9)		517(28.3)	1313(71.7)	
Secondary	1518(35.7)	2731(64.3)		21,938(18.7)	95,093(81.3)		1160(51.5)	1091(48.5)		952(42.5)	1289(57.5)		606(30.4)	1389(69.6)	
Higher	411(30.4)	939(69.6)		5498(15.2)	30,601(84.8)		607(46.1)	711(53.9)		164(33.5)	326(66.5)		62(28.8)	153(71.2)	
Father's education															
No education	435(33.4)	866(66.6)	0.136	892(16.8)	4429(83.2)	<0.003	1283(41.9)	1778(58.1)	<0.001	7823(42.7)	10,477(57.3)	<0.014	143(24.4)	442(75.6)	
Primary	1026(34.7)	1935(65.3)		905(19.4)	3750(80.6)		810(45.2)	983(54.8)		1981(43.0)	2625(57.0)		609(29.8)	1433(70.2)	
Secondary and	1485(34.2)	2854(65.8)		4553(18.2)	20,480(81.8)		2491(45.6)	2967(54.4)		3510(41.0)	5042(59.0)		677(28.1)	1735(71.9)	
Higher	4(14.3)	24(85.7)		35(22.6)	120(77.4)		6(21.4)	22(78.6)		93(37.2)	157(62.8)		30(30.0)	70(70.0)	
Don't know															
Mass media exposure															
Not Exposed	1384(34.8)	2598(65.2)	0.313	20,755(18.0)	94,384(82.0)	<0.001	2261(42.1)	3114(57.9)	<0.001	6405(42.2)	8770(57.8)	0.874	776(27.2)	2079(72.8)	
Exposed	1616(33.7)	3175(66.3)		20,149(17.4)	95,583(82.6)		2404(47.0)	2716(53.0)		7032(42.3)	9594(57.7)		695(29.7)	1643(70.3)	
Number of ANC															
No ANC	168(41.5)	237(58.5)	0.119	3051(23.5)	9905(76.5)	<0.001	374(44.3)	470(55.7)	<0.001	3199(40.7)	4655(59.3)	<0.001	20(26.9)	55(73.3)	
Less than 8	1599(39.1)	2488(60.9)		25,148(19.6)	103,185(80.4)		2504(50.8)	2425(49.2)		5954(52.7)	5353(47.3)		814(31.3)	1789(68.7)	
8+	197(35.3)	361(64.7)		6563(19.5)	27,095(80.5)		544(58.1)	393(41.9)		270(57.3)	201(42.7)		66(33.3)	110(66.7)	

Table 3 (continued)

Variables/ Categories	Child morbidity						Afghanistan						Nepal		
	Bangladesh			India			Pakistan			Yes			No		
	No N (%)	P-value	Yes N (%)	No N (%)	P-value	Yes N (%)	No N (%)	P-value	Yes N (%)	No N (%)	P-value	Yes N (%)	No N (%)	P-value	
Delivery by caesarean section															
No	1378(38.4)	2206(61.6)	0.637	31,715(17.5)	149,521(82.5)	<0.001	3539(43.5)	4604(56.5)	<0.001	13,000(42.1)	17,864(57.9)	<0.001	735(29.4)	1764(70.6)	0.074
Yes	660(37.8)	1087(62.2)		9188(18.5)	40,446(81.5)		1116(47.7)	1223(52.3)		422(49.0)	439(51.0)		190(33.2)	382(66.8)	
Place of delivery															
Respondents Home	1051(39.4)	1619(60.6)	0.081	4647(18.0)	21,201(82.0)	0.243	1475(41.7)	2058(58.3)	<0.001	6917(42.7)	9271(57.3)	0.245	166(27.3)	443(72.7)	0.089
With Health Facility	988(37.0)	1679(63.0)		36,256(17.7)	168,765(82.3)		3190(45.8)	3768(54.2)		6515(42.1)	8966(57.9)		758(30.8)	1704(69.2)	
Mother's BMI															
Underweight	421(35.9)	753(64.1)	<0.030	8161(19.7)	33,361(80.3)	<0.001	182(44.6)	226(55.4)	0.330	N/A			94(27.2)	251(72.8)	0.159
Normal	1734(35.5)	3152(64.5)		22,028(17.6)	103,267(82.4)		723(48.3)	774(51.7)					443(29.2)	1076(70.8)	
Overweight/obese	698(32.4)	1456(67.6)		6769(16.7)	33,844(83.3)		688(46.3)	797(53.7)					208(32.6)	431(67.4)	
Sex of household head															
Male	2582(34.0)	5017(66.0)	0.266	34,098(17.4)	161,730(82.6)	<0.001	4130(44.2)	5204(55.8)	0.236	13,279(42.2)	18,168(57.8)	0.362	1013(28.6)	2535(71.4)	0.575
Female	418(35.6)	755(64.4)		6805(19.4)	28,235(80.6)		535(46.1)	626(53.9)		158(44.6)	196(55.4)		457(27.8)	1187(72.2)	
Wealth index															
Poor	1276(34.8)	2388(65.2)	<0.035	20,388(19.1)	86,552(80.9)	<0.001	1899(42.3)	2587(57.7)	<0.001	5289(41.9)	7343(58.1)	0.526	646(27.5)	1700(72.5)	0.132
Middle	593(36.0)	1053(64.0)		8233(18.3)	36,868(81.7)		1011(46.4)	1167(53.6)		2886(42.6)	3892(57.4)		332(30.8)	747(69.2)	
Rich	1131(32.7)	2331(67.3)		12,282(15.6)	66,546(84.4)		1755(45.8)	2076(54.2)		5262(42.5)	7129(57.5)		493(27.9)	1274(72.1)	
Place of residence															
Urban	769(31.9)	1644(68.1)	<0.005	9922(16.1)	51,606(83.9)	<0.001	1533(45.7)	1819(54.3)	0.070	3269(45.1)	3977(54.9)	<0.001	991(29.4)	2376(70.6)	<0.016
Rural	2231(35.1)	4129(64.9)		30,981(18.3)	138,361(81.7)		3132(43.8)	4011(56.2)		10,168(41.4)	14,387(58.6)		480(26.3)	1346(73.7)	

Note: Table 3 shows the bivariate associations between early maternal marriage, socioeconomic and demographic characteristics, and morbidity among children under the age of five. Pearson's Chi-squared test was used to analyze the connections. A p-value of less than 0.05 indicates statistical significance

N/A= Not Available

Table 4 Adjusted odds ratios (AOR) and 95% confidence intervals for under-five morbidity by early marriage and socioeconomic factors in five South Asian countries

Variables/Categories		Child morbidity																			
		Bangladesh				India				Pakistan				Afghanistan				Nepal			
		95% CI	P-value	ORs	95% CI	upper-lower	P-value	ORs	95% CI	upper-lower	P-value	ORs	95% CI	upper-lower	P-value	ORs	95% CI	upper-lower	P-value		
Age at first marriage	Early marriage	1	1	0.145	1	0.82***	1	0.000	1	0.075	1	0.000	1	0.939							
	Adult marriage	0.90	0.78–1.04				0.77–0.88		0.83	0.67–1.02	0.87***	0.82–0.92	0.99	0.76–1.30							
Child sex	Male	1	1	0.102	1	0.92*	1	0.013	1	0.000	1	0.334	1	0.909							
	Female	0.91	0.81–1.02				0.87–0.98		1.43***	1.20–1.70	0.97	0.92–1.03	1.01	0.80–1.28							
Child twin	Single birth	1	1	0.712	1	0.84	1	0.316	1	0.132	1	0.001	1	0.671							
	Multiple births	1.12	0.62–2.00	62.00			0.59–1.20		0.58	0.29–1.18	0.62***	0.47–0.83	0.69	0.13–3.79							
Birth order	First	1	1	0.287	1		1	0.005	1	0.327	1	0.505	1	0.321							
	Second or third	1.08	0.94–1.25			0.90**	0.84–0.97	0.000	1.14	0.88–1.47	0.97	0.88–1.06	1.13	0.88–1.50	0.473						
	Fourth and up	1.16	0.93–1.45			0.76***	0.68–0.86		1.10	0.84–1.44	0.93	0.85–1.02	0.84	0.52–1.36							
Currently breastfeeding	No	1	1	0.003	1	1.32***	1	0.000	1	1	1.33–1.91	1.07*	1	0.131							
	Yes	1.29***	1.09–1.52				1.24–1.42		1.59***	1.09–4.37	1.22*	1.01–1.13	1.44	0.90–2.32							
Mother's age	15–18	1	1	0.696	1	0.98	1	0.855	1	0.028	1	0.046	1	0.608							
	19–49	0.96	0.77–1.19				0.76–1.25		2.18*	1.09–4.37	1.22*	1.01–1.49	0.85	0.45–1.59							
Maternal education	No education	1	1	0.252	1		1	0.217	1	0.029	1	0.013	1	0.870							
	Primary	0.86	0.66–1.12			1.08	0.96–1.22	0.029	1.37*	1.03–1.82	0.000	1.15*	1.03–1.28	0.937	1.03	0.70–1.52	0.407				
	Secondary	0.91	0.69–1.19			1.12*	1.01–1.24	0.003	1.88***	1.40–2.53	0.096	0.97	0.86–1.09	0.000	1.20	0.78–1.87	0.071				
Father's education	Higher	0.79	0.57–1.08			0.81***	0.70–0.93		1.35	0.95–1.93	0.57***	0.45–0.72	1.85	0.95–3.59							
Mass media exposure	No education	1	1	0.675	1		1	0.325	1	0.595	1	0.637	1	0.616							
	Primary	1.04	0.86–1.27			1.07	0.94–1.21	0.009	0.93	0.70–1.23	0.086	0.98	0.90–1.07	0.479	0.90	0.55–1.42	0.983				
	Secondary and Higher	1.04	0.84–1.27			1.16**	1.04–1.29	0.220	0.80	0.63–1.03	0.839	0.97	0.91–1.05	0.099	1.01	0.61–1.67	0.612				
Delivery by caesarean section	Don't know	0.28	0.06–1.23			1.33	0.85–2.08		1.24	0.16–9.58	0.76	0.55–1.05	1.26	0.52–3.06							
Number of ANC	Not Exposed	1	1	0.390	1	1.08*	1	0.047	1	0.099	1	1	0.675	1	0.346						
	Exposed	1.06	0.93–1.22				1.01–1.16		1.19	0.97–1.46	0.99	0.93–1.05	0.90	0.70–1.14							
Number of ANC	No ANC	1	1	0.881	1		1	0.010	1	0.008	1	0.000	1	0.770							
	less than 8	0.98	0.78–1.23			0.85**	0.75–0.96	0.014	1.51**	1.12–2.05	0.007	1.73***	1.63–1.85	0.000	1.13	0.51–2.49	0.601				
	8+	0.89	0.66–1.19			0.84*	0.73–0.97		1.75**	1.17–2.63	2.23***	1.83–2.71	1.28	0.51–3.21							

Table 4 (continued)

Variables/Categories	Child morbidity												
	Bangladesh			India			Pakistan			Afghanistan			Nepal
No	1	1	0.369	1	0.028	1	0.210	1	1	0.063	1	1	0.644
Yes	1.07	0.91–1.30	1.09*	1.01–1.18	0.86	0.69–1.09	1.17	0.99–1.38	1.08	0.79–1.47			
Place of delivery	1	1	0.478	1	0.453	1	0.005	1	0.000	1	0.745		
	0.94	0.79–1.12	0.96	0.86–1.07	0.72**	0.57–0.90	0.78***	0.73–0.83	1.06	0.76–1.48			
Mother's BMI													
Underweight	1	1	0.303	1	0.000	1	0.327	N/A	1	1	0.295		
Normal	0.92	0.78–1.08	0.084	0.86***	0.80–0.93	0.009	0.86–1.56	0.979	1.20	0.85–1.69	0.863		
Overweight/obese	0.83	0.68–1.03	0.87**	0.78–0.97	1.01	0.74–1.37			0.96	0.64–1.45			
Sex of household head													
Male	1	1	0.970	1	0.003	1	0.716	1	0.715	1	0.272		
Female	1.01	0.84–1.20	1.14***	1.05–1.24	0.95	0.71–1.27	1.05	0.81–1.36	0.87	0.67–1.12			
Wealth index													
Poor	1	1	0.586	1	0.014	1	0.875	1	0.096	1	0.750		
Middle	1.05	0.88–1.25	0.691	0.89*	0.82–0.98	0.000	0.75–1.28	0.286	0.94	0.69–1.31	0.466		
Rich	0.97	0.81–1.15	0.79***	0.72–0.87	0.85	0.62–1.15	0.97	0.90–1.06	0.89	0.66–1.21			
Place of residence													
Urban	1	1	0.286	1	0.529	1	0.674	1	0.000	1	0.803		
Rural	1.09	0.93–1.26	1.03	0.95–1.12	0.95	0.77–1.19	0.80***	0.73–0.87	0.97	0.75–1.25			

Note: * Adjusted odds ratios (AORs) with 95% confidence intervals (CIs) are displayed. Each model is stratified by nation and includes predictors at the child, mother, and household levels. The table indicates reference

Note: * Adjusted odds ratios (AORs) with 95% confidence intervals (CIs) are displayed. Each model is stratified by nation and includes predictors at the child, mother, and household levels. The table indicates reference categories

AORs with *p < 0.05, **p < 0.01, and ***p < 0.001 are statistically significant. 1= Reference category, and N/A= Not Available

association with under-five morbidity in India and Afghanistan; However, the direction of the association indicates lower odds in some models. This unexpected pattern may reflect residual confounding, selective survival, or differential reporting behaviors among younger mothers rather than a true protective effect, as previous research has demonstrated that early marriage affects maternal autonomy and access to child health services [5, 19]. The early married child was significantly associated with an increased likelihood of under-five child morbidity than the adult married child [27]. In India, female children were associated with a lower risk of morbidity, but in Pakistan, female children had greater odds, indicating that sex-based disparities in treatment or disease vulnerability differ by environment. It explicitly differentiates morbidity rates between boys and girls [1, 5, 6, 8]. In addition, multiple births were linked with significantly decreased morbidity in Afghanistan. Multiple births are substantially linked to greater rates of low birth weight and preterm birth, two factors that considerably exacerbate these children's morbidity [28, 29]. Birth order appears to have a strong link to the parental treatment of child morbidity, especially when second- and third-born born are compared by child morbidity. Based on birth order, parental health investments varied significantly [30, 31]. Interestingly, children who were currently breastfeeding showed significantly higher odds of morbidity across most countries. This paradoxical finding likely reflects reverse causality, whereby mothers continue or increase breastfeeding in response to child illness, rather than breastfeeding causing illness. Similar patterns have been documented in previous studies [32]. Breastfeeding, especially exclusive breastfeeding, is linked to infant morbidity and infection risks [33–35]. Maternal age yielded conflicting results. In Pakistan, older women (19–49) were substantially more likely to report child morbidity, whereas in Afghanistan, the increase was small. These findings may reflect aging-related differences in birth spacing, health literacy, or maternal weariness. Other research found similar connections [36]. Maternal education was significantly beneficial in India and Afghanistan, according to global evidence that educated mothers are more likely to participate in effective health-seeking and disease-preventive behaviors [37]. In contrast, paternal education was only substantially linked with morbidity in India, which could be attributed to better health awareness and reporting. This emphasizes how crucial elements like access to healthcare are [38]. However, depending on additional variables like geographic location and economic standing, the effect of schooling on child morbidity can differ [39, 40]. The habit of child feeding may benefit from the presence of educated parents [41]. In India, mass media exposure was related to increased morbidity, which may

indicate that media-literate mothers are more likely to notice and report disease signs [42, 43]. Antenatal care (ANC) had a wide range of associated. The association between antenatal care (ANC) visits and child morbidity varied across countries. While higher ANC utilization was protective in India, increased ANC visits in Pakistan and Afghanistan were associated with higher reported morbidity. This may indicate that women with high-risk pregnancies or existing child health concerns are more likely to seek repeated ANC services, or it may reflect variations in the quality of care rather than quantity alone [44]. The best maternal and fetal outcomes are associated with antenatal care (ANC) [45, 46]. Institutional delivery significantly reduced morbidity in Pakistan and Afghanistan, highlighting the importance of professional medical care at birth in mitigating early childhood disease risks. The immune system may develop differently in babies born with CS, increasing the risk of allergies, asthma, and a less diverse gut microbiota [47, 48]. Children of mothers who were born in Health facilities were less likely to have child morbidity [49]. Maternal BMI (normal range) was related to lower morbidity only in India, which may reflect the country's nutrition transition and expanding urban-rural dietary disparities [50]. Underweight mothers have a higher chance of having low birth weight and small-for-gestational-age babies, which are connected to higher morbidity [51, 52]. Premature birth, newborn asphyxia, fetal overgrowth, and increased neonatal morbidity are among the negative outcomes that are linked to higher maternal BMI [53, 54]. Additionally, from households where the household head was female were more likely to have child morbidity [55]. Wealth index likewise had a high protective impact in India, but the link was modest or nonsignificant in Bangladesh, Pakistan, and Nepal. This emphasizes the importance of taking into account country-specific health financing and access limits when understanding the associated effects on health outcomes [56, 57]. Finally, rural living was unexpectedly protective in Afghanistan, which could be attributed to reduced disease reporting, less environmental exposure, or changes in health-seeking behaviors. There was no consistent correlation in the other countries [58, 59]. Taken together, our findings demonstrate that early marriage is a significant contextual predictor of under-five morbidity, although its effects are frequently moderated or changed by maternal education, healthcare utilization, and household wealth. The study emphasizes the need for multisectoral measures, such as health system strengthening, education promotion, and gender equity reforms, in improving child health outcomes in South Asia. The surprising direction of several correlations found in this study may be partially explained by unmeasured confounders, selective survival of healthier children, and variations in health-seeking and reporting habits among

socioeconomic groups. Sensitivity analysis, such as stratified models or propensity score adjustment, may be used in future studies to evaluate the robustness of these results.

Strengths and limitations

This study has various advantages, including a comprehensive cross-country analysis employing nationally representative data, which improves the generalizability and validity of conclusions using robust statistical approaches. It contributes to the current literature by emphasizing the importance of maternal education and prenatal care in child health. However, a major limitation of this study is its cross-sectional design, which precludes any causal inference between early marriage and child morbidity. The observed associations should therefore be interpreted as correlations rather than causal relationships. In addition, self-reported morbidity data may be subject to recall and reporting bias, particularly across different cultural contexts.

Conclusion

This study highlights that early marriage and socioeconomic disadvantage are strongly associated with under-five morbidity in South Asia. These findings underscore the urgent need for multisectoral interventions, including strengthening girls' education, enforcing laws to delay age at marriage, expanding community awareness programs, and improving the quality and accessibility of maternal and child healthcare services, particularly antenatal care. Policymakers should prioritize integrated strategies that address both social norms and health system gaps to reduce childhood morbidity and improve child survival in the region. Future studies should investigate region-specific trends and community-based measures to mitigate the effects of early marriage on child health outcomes.

Abbreviations

DHS	Demographic and Health Survey
BDHS	Bangladesh Demographic and Health Survey
NFHS	National Family Health Survey
PDHS	Pakistan Demographic and Health Survey
AfDHS	Afghanistan Demographic and Health Survey
NDHS	Nepal Demographic and Health Survey
HIV	Human Immunodeficiency Virus
AIDS	Acquired Immunodeficiency Syndrome
ARI	Acute Respiratory Infection
ANC	Antenatal Care
BMI	Body Mass Index
SPSS	Statistical Package for the Social Sciences
MLE	Maximum Likelihood Estimation

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Authors' contributions

JH (Jakir Hossain) conceptualized the study, developed the main statistical analysis, drafted the manuscript, wrote the main script, and wrote the introduction. ASMR (Abu Sayeed Md. Ripon Rouf) reviewed the manuscript critically. MT (Muhammad Tareq) checked and proofread the final document. MR (Md. Rokunuzzaman) handled the data processing and management. SKDS (Samrat Kumar Dev Sharma) reviewed the final document and refined this. All authors read and approved the final manuscript.

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Data availability

The datasets used and/or analyzed during the current study are available from the DHS Program upon reasonable request and registration. Data access is granted at: <https://dhsprogram.com/data/available-datasets.cfm> (<https://dhsprogram.com/data/available-datasets.cfm>).

Declarations

Ethical approval and consent to participate

This study's foundation examined publicly accessible survey data sets stripped of all respondent-identifying information. The relevant Ethics Committees in Bangladesh, India, Pakistan, Afghanistan, Nepal, and ICF Macro at Calverton, USA, approved this survey. The Demographic and Health Surveys (DHS) program data archivist permitted the authors to download the dataset used in this investigation (<https://dhsprogram.com>).

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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